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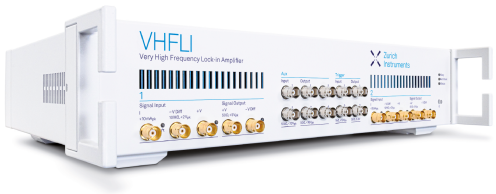

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Superconducting flux pumps

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ABSTRACT

This paper presents a perspective on flux pumps. Flux pumps are a means of providing current to a coil or a magnet without direct electrical or physical contact. Typically, a coil would be charged directly using current leads, which would then be withdrawn if the magnets operated in the persistent mode (as with magnetic resonance imaging) or left in place if the field required maintenance or ramping. Flux pumps use the principle of magnetic induction. They present two distinct advantages. The first is that there is no direct connection to the external world and, therefore, no associated heat loss. The second is that, although high current can be developed in the magnet or the coil, these can be produced without the need for high-current power supplies, which are bulky and expensive. Flux pumps are especially appropriate for machines constructed from high temperature superconductors (HTSs). They facilitate the full use of the HTS's current carrying capability, cheaply and simply, enabling smaller, lighter, and more powerful magnets, motors, and generators.

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I. INTRODUCTION

Until now, we have had magnetic resonance imaging (MRI) machines, typically made from low temperature superconductors (LTSs), which perform a vital function in the healthcare market. MRI machines are closely related to nuclear magnetic resonance (NMR) machines, which enable us to push back the boundaries of physics examining states of matter inaccessible by any other means. NMR machines are typically classed by the frequency at which they work. The higher the frequency, the better the resolution of the machine. Typically, NMR machines are available up to 900 MHz, which requires a flux density of 21.2 T, although Bruker markets the 1 GHz Aeon, which uses a flux density of 23.5 T. This is close to the H_{C2} of Nb_3Sn of 24.5 T and, therefore, a higher frequency NMR will require a different material. Higher field magnets do exist, but these typically use a combination of a resistive (bitter) magnet in combination with an LTS magnet. The highest field magnet currently is 45 T, which is sited at the National High Magnetic Field Lab (NHMFL) in Florida. However, a resistive magnet is expensive to run and requires a large amount of supporting infrastructure. Under development are hybrid magnets which use both HTS and LTS. The record is again held by the NHMFL that has constructed a 32 T magnet which uses a combination of three types of superconductors: high critical temperature (T_c) $YBa_2Cu_3O_{7-\delta}$ (YBCO) and low T_c niobium tin and niobium titanium. All of these figures though pale into insignificance compared to the maximum field which HTS can sustain, which is thought to be of the order of 100 T.

A perennial problem with magnets of any size is quenching. Quenching occurs when the stored energy of the magnet is suddenly released. Reducing the inductance of a magnet dramatically reduces the challenge of protecting the magnet from sudden quenches. Reducing the inductance also reduces charging times and voltages. For equivalent magnets, the inductance decreases as the square of the operating current increases, since increasing the operating current proportionally reduces the number of turns required to produce the same field, and inductance is a function of the number of turns squared.

Coated conductors have very large current carrying capabilities both individually and in the form of a cable. It is not inconceivable that a magnet could be constructed which uses 20 kA conductors. A magnet constructed with such a cable would, in comparison with the one constructed from 200 A conductors, have an inductance four orders of magnitude smaller, be easier to protect, and be easier to ramp to field due to the lower charging voltage.

For fields above 18 T, the engineering critical current density of the REBCO (rare-earth barium copper oxide) tape is higher than any other material. BSCCO (Bi-Sr-Ca-Cu-O compounds) also has a higher engineering current density than Nb_3Sn above about 16 T.

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FIG. 1. Current leads for the JT60-SA (F4E and JAEA representatives) with Stefan Kaufmann, Member of German Parliament (sixth from the left) in front of the delivered HTS-CLs. Reproduced with permission from Fusion for Energy—25 December 2015, https://f4e.europa.eu/mediacorner/newsview.aspx?content=965. Copyright 2015, European Joint Undertaking for ITER and the Development of Fusion Energy ("Fusion for Energy").

Thus, we can make a strong case for high-current high-field minimum inductance magnets. The question is what is the best way of achieving the high current.

There are only two practical methods of providing the current: directly, using current leads, or inductively, using flux pumps. The choice depends on a number of things. Traditionally, and especially for low T_c magnets where the magnet is running in the persistent mode, such as with an MRI, current leads are the obvious choice. LTS joints can be made which effectively have zero resistance. Thus, the magnet can be charged, the current leads withdrawn, and the field remains, more or less indefinitely.

For an HTS magnet, or a magnet in which the field needs to be regularly ramped, the situation changes. The magnet will need to be permanently connected to a power supply, and thus the current leads, if they are being used, will need to be permanently in place. This represents a thermal overhead, approximately 45 W/kA; also, if the current is especially high such as with the JT60-SA in which a 20 kA supply is used, the current leads themselves are extremely bulky (Fig. 1) and the required power supplies are expensive.

Alternatively, such as with those used in the LHC orbit corrector (Fig. 2), current leads can be tortuous, require multistage cooling, and will themselves dictate the current, which can be used.

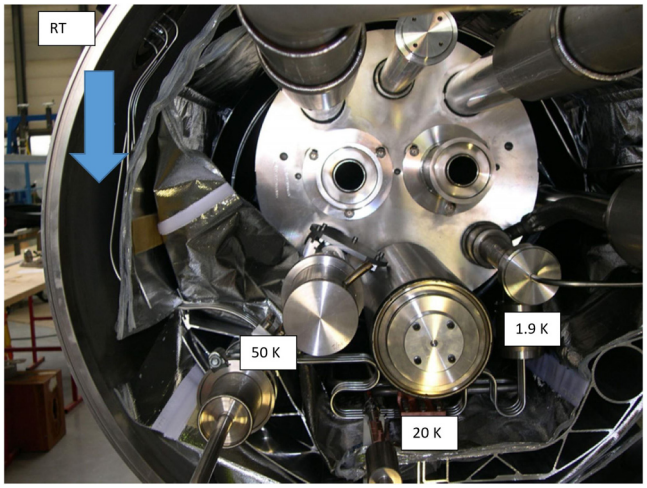

FIG. 2. Current leads, links, and buses. The current leads enter at the top left at room temperature (RT and blue arrow) pass through stages at 50 K and 20 K before finally arriving at the magnet at 1.9 K. Reproduced with permission from Ballarino, e-print arXiv:1501.07166v1. Copyright 2015, A. Ballarino.

II. DISCUSSION

Flux pumping represents a low cost and compact alternative to the use of current leads. Originally conceived decades ago (in the 1960s), the principle as summarized in the review by van de Klundert *et al.*[3] was applied to low temperature superconductors, and the principle was that by creating superconducting switches, flux could be transferred into a magnet. The principle is straightforward: since a superconducting ring will prevent the passage of flux either into or out of the ring, the use of two connected superconducting circuits, one of which contains the device to be magnetized and both containing switches, will allow flux to be transferred to the device to be magnetized. There are many variations which were summarized for LTS flux pumps by van de Klundert *et al.* and for HTS flux pumps by Coombs *et al.*[4] They can be broadly classified into flux pumps which rely on the interruption of superconductivity and those which do not.

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A. Flux pumps which rely on the interruption of superconductivity

In Fig. 1 of his paper (reproduced as Fig. 3), van de Klundert illustrates the principle. Switch S1 is first opened by breaking superconductivity in this section. Flux is transferred to the loop. Switch S1 is then closed and switch S2 opened allowing flux to the load L. Switch S2 is then closed and the process repeated, incrementally charging the load.

Included in van de Klundert's references is one to a paper written by Atherton and published in Cryogenics in 1967. In this short note, he points out that as long as the switches are opened completely and that consequently there is no current through the switches, the theoretical efficiency is 100%. In practice, this limit is not practically reachable as the normal way of operating the switches is to heat them rather than physically open them, the resistance of the switch is, therefore, finite. Nevertheless, by reducing the current through the switches before they are opened, the overall efficiency can be increased. The highest current achieved at the time of Van de Klundert's review was 8 kA reported by Spiel and Boom.[5]

Then, in 1991, Mulder *et al.* built and tested a thermally switched flux pump that was capable of providing a power of 30 W and a current of 100 kA.[8,9] Although it was only tested up to 25 kA, this represents the highest current so far from a flux pump. The pump was operated at relatively low frequency but a later paper from the same group demonstrated that a lower current capacity pump (300 A) albeit with a higher power (100 W) could be operated at main frequencies.[10] Additional work reported in 2002 by Jeong *et al.*[7] has recently been followed up by the same group in Ref. 8, where HTS BSCCO coil shims are energized by a Nb_3Sn flux pump.

All of the above devices were based on LTS switches and the typical mechanism was the use of heat to open the switch. They have high efficiencies, typically upward of 80% and in the case of Mulder *et al.* and Shevchenko greater than 96% during pumping and 98% during the current maintenance mode.

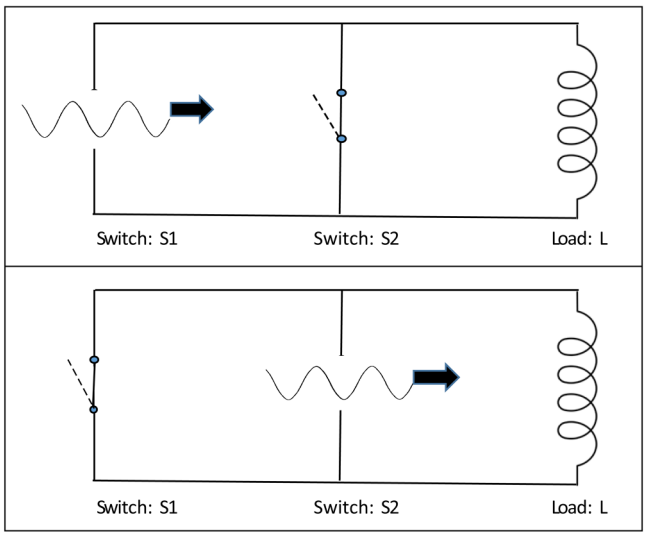

FIG. 3. Flux pump circuit. Figure is based on Fig. 1 of van de Klundert and ten Kate, Cryogenics **21**(4), 195–206 (1981).

Thus, it is clear from the low Tc studies that a flux pump can provide high currents at high efficiencies and has great potential for charging a magnet or other devices. In general, these flux pumps share a common characteristic, that is, they require superconductivity to be interrupted in order to develop impedance and hence a switch.

There are two problems which can occur when using a bridge or a switch which require the loss of superconductivity. The first is that of necessity the bridge must be allowed to cool between activations. It is notable that an LTS pump has been developed which works at main frequency but that of necessity was a relatively low current device. The kA class devices work at a much lower frequency. The reason why a higher frequency is desirable is primarily due to the ripple field. The longer the cycle, the greater the decay between charging and hence, the greater the magnitude of the ripple. Oomen *et al.*[6] created a flux pump using HTS thin films in which he used heat to switch the superconductor. The operation speed was relatively low at 0.1 Hz, and as a result, the flux ripple was approximately 2400 ppm. In the same paper, he substituted MOSFETS for the HTS switches, raised the operating frequency to 50 Hz, and improved the flux ripple by some two orders of magnitude to 60 ppm. Note if you simply divide the flux ripple by the change in the frequency, this would indicate a flux ripple of 4.8 ppm. The difference is caused by the characteristics of the MOSFETS themselves and is primarily due to the fact that when conducting they do not have zero resistance.

The second is a more interesting problem and is the potential requirement for active and accurate control. As soon as superconductivity is lost, the resistance of the device starts to rise. This is the principle that is exploited in fault current limiters. Early development of fault current limiters omitted the need to provide an alternative path for the current to mitigate the temperature rise and thereby prevent the device burning out. Once this was realized, in HTS fault current limiters at least, a copper stabilizer layer was added. This approach could be taken with an HTS bridge to obviate the need for control, but then the impedance of the bridge would be limited to that of the stabilizer layer. Thus, the bridge would, of necessity, be very long in order to generate a sizable voltage across the load.

B. Flux pumps which do not rely on the interruption of superconductivity

Recent research on high temperature superconductors has made it clear that it was not necessary to drive the superconductor normal. In 2008, Coombs *et al.*[11] proposed the traveling wave flux pump in which field is moved across a superconductor, and then in 2015, the same group proposed the use of the dynamic voltage to realize the HTS rectifier type flux pump.[12] Other groups include Hoffmann *et al.* who demonstrated that the traveling wave principle could be applied to pumping a coil.[13]

Numerous groups have tried to explain the traveling wave flux pump[14–17] and many groups are trying to use it to activate a motor or a generator rotor coil, where the motion of the rotor can be used to develop the traveling wave.[18,19]

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The HTS rectifier type flux pump is easier to explain and its operation comes from the fact that if a tape is carrying a transport current, then an oscillating field applied perpendicular to its surface will induce a voltage, and that the voltage (although changing) will always have the same sign. This effect is called the dynamic voltage and there are numerous references in the literature, but a particularly good one can be found in Ref. 20.

Neither the traveling wave nor the HTS rectifier type of flux pump require the interruption of superconductivity. This is an important distinction from all previous types of flux pumps. It enables the flux pump to be operated at a temperature far removed from its transition temperature and thus to better utilize the current carrying capabilities of the superconductor.

It is important to recognize that in both types of HTS pump (traveling wave or HTS rectifier), the active element, the bridge, or switch must be capable of carrying a higher, current than that carried by the load. To achieve this, the bridge could employ more conductors than the load or be operated at a lower temperature. Alternatively, since the bridge will see lower fields than the load, it will, even if it is made from a similar amount of conductor and is operated at the same temperature, have a higher current carrying capability than the superconductor in the load.

However, two things must be born in mind. The first is that, as can be seen from Fig. 1, the field dependence of Jc in YBCO, in particular, is not particularly high, especially so when the field is parallel to the plane of the conductor. The second is that if more conductors are used in the bridge element, there will be the problem of making the connection to the load. Jointing YBCO is not particularly easy and although techniques are being developed to join multiple tapes,[21,22] this adds unnecessary complication and is best avoided as it inevitably leads to higher resistance when either connecting the superconductor to the substrate or alternatively crossing tapes.

Although the mechanisms that support the traveling wave and the HTS rectifier type flux pumps are similar in terms of the physics which underpin them, there is a fundamental difference. In the traveling wave type flux pump, everything is achieved with a single loop. That is, the load is short circuited by a single section of superconductor to which the traveling wave is applied. This means that the current through the load and the current through the pump section are always similar. This is a fundamental weakness as it means that the losses cannot be minimized by reducing the current through the bridge section as they can with the rectifier type. Thus, although the traveling wave type is the simplest form of all, uses the least superconductor, and does not require a transformer, it will always suffer from greater losses and ultimately this will limit its operation to lower currents than can be achieved with a bridge rectifier type.

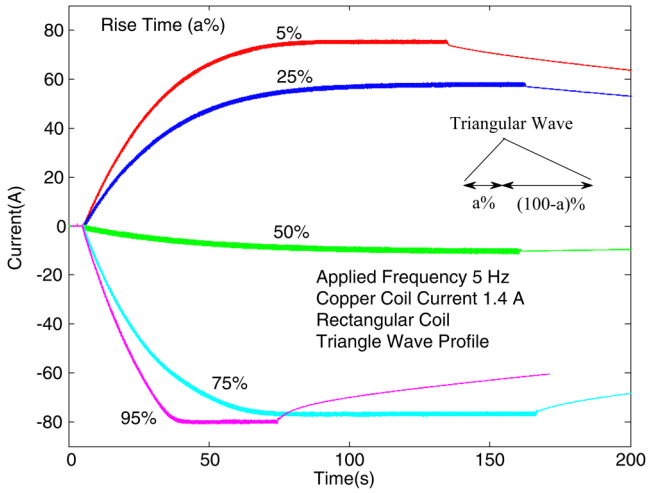

FIG. 4. Effect of changing the duty cycle on the sign of current induced by a triangular traveling wave.[23]

Flux pumps of all types can be operated in both rectifying and inverting modes. This is self-evident in the bridge rectifier types and the rotating magnet types (simply reverse the direction of rotation). For the traveling wave type, changes in the wave form can also bring about the same effect (Fig. 4) regardless of the direction of travel.

Thus, they can be used for close control of the current in the load and, in principle at least, be used in energy storage devices such as Superconducting Magnetic Energy Storage (SMES). It is unlikely however that they would be used in SMES as their operation is relatively slow and the principle advantage of SMES would be for rapid discharge.[24]

III. CHARACTERISTICS

Flux pumping has been known about for many years. Impressive work has been carried out on low temperature flux pumps and devices with both high currents and high powers were developed. To date though, these have had relatively little commercial success. This is due to several possible factors, which apply specifically to low Tc magnets. The factors are

Operating in persistent mode

- It is possible to connect low temperature superconductors with low or no resistance joints. Hence, magnets (and MRI in particular) can be operated in the persistent mode without the need for an external power supply. A simple thermally operated superconducting switch together with retractable current leads can be used to charge them. This obviates the need for a device such as a flux pump.

Operating in ramped mode

- High temperature superconducting current leads now provide a low loss means of connecting the magnet to a power supply.

Operational stability

- Flux pumps, although they are "DC" supplies, are not constant current supplies. That is, they are pulsed. Traditional flux pumps, developed for use with LTS typically require heating the superconductor through its transition point and then allowing it to cool again. This limits the frequency at which they can be operated and hence whether they can be used for devices such as MRI or NMR. 50 Hz supplies were created but these were only able to provide low current limiting their use.

When we consider high Tc magnets, another picture emerges.

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Operating in persistent mode

- Although it is now possible to make low resistance joints to date, it is not practical to make zero resistance joints. This means that the magnet needs to remain connected to the power supply. Although there is still the option of using current leads, this represents a thermal overhead and if a high current is required and current leads are used a space overhead.

Operating in ramped mode

- Flux pumps can be built to provide any required power. Thus, in principle, a magnet powered by a flux pump can be powered up and down as required.
- High currents. HTS can sustain very high currents in an extremely compact space and thus the means to provide the highest possible current is a strong advantage. The cost of an HTS flux pump will scale with the power rather than the current meaning that a kiloampere flux-pumped power supply is potentially much cheaper than the equivalent conventional power supply.

Operational stability

- HTS flux pumps, whether they are traveling wave type or transformer-rectifier type, do not require the interruption of superconductivity. Thus, they can be operated at sufficiently high frequencies to provide a power supply with a stability of 1 ppm or more. Importantly, also, transformer-rectifier types can be tuned to provide a variable charging voltage, as the transformer

TABLE I. Comparison between major variants of flux pumps.

Method	Characteristics	Pros	Cons	Notes
Moving normal spot	Flux is drawn into a superconducting circuit via a moving normal spot	Possibly the simplest method. A normal spot is induced in the superconductor, which contains fluxons. As the spot moves across the superconductor, the fluxons move with it	Of limited use. Both the source and the spot must move together	This method requires only single spot
Thermally induced switches	Flux is permitted to enter a superconducting circuit by the removal of a section of the superconducting circuit (typically by heating)	The spot is widened to include the whole width of a section of superconductor. Flux can then pass through this section. A second switch allows the flux to move to the load. The switches are operated sequentially	Superconductor must be heated to its transition point. This limits the frequency of operation of the device	Low frequency operation leads to decay between cycles and hence a higher ripple field. 50 Hz operation has been achieved but only at relatively low currents
HTS traveling wave	Flux flow in the superconductor creates areas of different impedance	Superconductivity is never destroyed (in principle at least). No requirement for heating	The system is not lossless. A loss occurs due the motion of the flux and the development of the charging voltage. Also, the current in the load cannot be separated from the current through the switch. Thus, the switch impedance is not independent of the voltage developed	
HTS rectifier type	Flux traverses into a superconducting circuit when the superconductor is full of current and hence unable to resist the incursion	In this system, the current through the switch can be controlled independently of the load. There is no requirement for heating since superconductivity is never destroyed. Hence, the system can operate at high frequencies	The system is not lossless. A loss occurs due to the motion of the flux and the development of the charging voltage	
Power electronic switches	In essence, this is not a flux pump although the operation is similar. It is simply a half or full wave rectifier	MOSFETs (or similar) are readily available for high currrents and have high switching speed	MOSFETS in common with all power electronics have on state resistance.	Ripple is large due to on state resistance of MOSFETs

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and the bridge are operated independently which cannot be done in a rotating magnet type (Table I).

IV. APPLICATIONS

There are many potential applications for HTS machines and flux-pumped HTS machines, in particular. Here, we will highlight three.

- Motors/generators: Flux pumps have been proposed for use in motors/generators, and there are various research projects around the world looking at gearing the traveling wave effect to the charging of motors. In principle, at least, very high power density machines (in terms of kilowatt per kilogram) can be made using superconductors, potentially enabling all electric wide-bodied jets and multimegawatt class windmills.
- Researchers in Victoria University of Wellington developed an exciter for HTS rotor windings.[25,26] They also achieved through-wall operation of the flux pump,[27] which for the first time moves the flux pump outside the cryogenic system. There is work going on in China and Korea too and recently a Korean group reported on a Wind Generator Design,[28] this work is interesting because it gives a direct comparison between current leads and flux pumps in their configuration. In their paper which is a design of a 12 MW wind turbine, they calculated that the heat load using a flux pump is 10.2 W and that using current leads is 31.8 W. This resulted in a slightly lower running temperature of 19.3 K vs 19.8 K. An additional advantage, not highlighted in their paper, is that the use of a flux pump obviates the need for brushes which are an expensive potential source of failure.
- High-field research magnets: Here, there is a very clear advantage for using HTS. LTS on its own cannot be used at fields greater than ca. 21 T and a great deal of work has been done looking at extending that range using HTS inserts. The highest field research magnet is 45 T which is in Florida at the NHMFL. It is a Bitter magnet but, being resistive, it is very expensive to run and requires a large amount of cooling and high power expensive power supplies. HTS has the potential to far surpass 45 T and a flux-pumped power supply may be the key to realizing this ambition.
- MRI: LTS MRI is a mature market. The magnets are relatively cheap (in terms of conductors). Consequently, it is unlikely that HTS will ever be competitive with LTS for whole body MRI. However, HTS with its high currents means that smaller devices which can be brought to the patient could have great utility. Example uses would be small animal imaging,[29] imaging limbs or importantly cephalic imaging.
- HTS MRI is to become a reality then of major concern is the stability of the power supply. To be of full utility, an MRI magnet requires the field to be maintained to 1 ppm. Tables II–IV summarize a calculation for stability and power requirements for a 3 T MRI magnet designed for cephalic imaging. The magnet bore and length are both 0.6 m and calculations are given for three operating currents 100 A, 1000 A, and 10 000 A. The calculation assumes that the longest practical length of superconductor available is 150 m and that the resistance of each joint is 20 nΩ. The stability criteria used is that the field has to be maintained within limits of 1 ppm.

Table II shows the overall statistics for the magnet itself for the three different configurations. Unsurprisingly, the mass and length of the conductor scale with the operating current due to the increased insulation required for each turn. Although the mass of the cryostat and the formers, onto which the tapes are wound, are not given here, the mass of the formers and their supports will reduce with the mass of the conductor. There is less weight to support and the electromagnetic forces are a function of the product of the current and the length of the conductor so are identical for all three magnets.

Table III details the requirements during charging and gives the time taken to charge and the power required vs three different charging voltages.

Table IV shows what is required to satisfy the stability requirements for the three different magnet currents. The important figure

TABLE II. 3 T MRI magnet overall characteristics.

	Operating current (A)		
	100	1000	10 000
Total turns required	23 878	2388	239
Conductor length (m)	45 000	4500	450
Conductor mass (kg)	500	50	5
Total no. of joints	302	32	5
Total coil resistance	7×10^{-6}	6.4×10^{-7}	1×10^{-7}
Inductance (H)	337.5	3.375	0.033 75
Time taken to decay by 1 ppm (s)	55.9	5.3	0.34
Stored energy (MJ)	1.69	1.69	1.69

TABLE III. 3 T MRI magnet charging requirements.

	Operating current (A)		
	100	1000	10 000
Charging voltage (V)	Charging time (h)		
1	9.38	0.938	0.0938
0.1	93.9	9.39	0.941
0.01	952	95.4	9.77
	Power supply (W)—assuming 90% efficiency		
1	56	560	5600
0.1	6	56	560
0.01	1	6	53

TABLE IV. 3 T MRI magnet requirements during operation.

	Operating current (A)		
	100	1000	10 000
Average voltage required to maintain field to 1 ppm (mV)	0.6	0.64	1.0
Frequency of pumping required to maintain field (Hz)	0.018	0.19	3.0

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here is the frequency required even though it is relatively low 3 Hz would be hard to achieve with HTS if it were required to be heated during each cycle, thus the rectifier type appears to be the most practical option for realizing a flux-pumped HTS MRI.

V. CONCLUSION

The development of high temperature superconductors capable of carrying extremely high currents in limited spaces should enable a sea-change in high-field/high-current electrical machines. However, to date, the impact of HTS has been relatively limited. This is partly due to the relative "newness" of the material and the fact that its primary use is in large machines, which require many man-years of development. Additionally, it has a high unit cost and to fully utilize its potential requires high-current/highly stable power supplies. An HTS flux pump is a low cost/low loss method of providing the current together with the stability required for high precision applications such as MRI and NMR. In addition, the traveling wave type has obvious benefits in a wide range of forms of rotating machinery such as windmills and high power density motors. Traveling wave and rectifier type flux pumps, while still under development, have, in the author's opinion, the potential to at least assist in bringing about the "HTS revolution" which we have been promised for many years.

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